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United States Patent	12400991
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Choi; Yun Hwa

Clip structure for semiconductor package and semiconductor package including the same

Abstract

Provided is a clip structure for a semiconductor package comprising: a first bonding unit bonded to a terminal part of an upper surface or a lower surface of a semiconductor device by using a conductive adhesive interposed therebetween, a main connecting unit which is extended and bent from the first bonding unit, a second bonding unit having an upper surface higher than the upper surface of the first bonding unit, an elastic unit elastically connected between the main connecting unit and one end of the second bonding unit, and a supporting unit bent and extended from the other end of the second bonding unit toward the main connecting unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit and has an elastic structure so that push-stress applying to the semiconductor device while molding may be dispersed.

Inventors:	Choi; Yun Hwa (Bucheon-si, KR)
Applicant:	JMJ Korea Co., Ltd. (Bucheon-si, KR)
Family ID:	1000008778671
Assignee:	JMJ Korea Co., Ltd. (Bucheon-si, KR)
Appl. No.:	17/839479
Filed:	June 14, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220399300 A1	Dec. 15, 2022

Foreign Application Priority Data

KR	10-2021-0076510	Jun. 14, 2021
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Publication Classification

Int. Cl.: H01L23/00 (20060101); H01L25/07 (20060101); H01L23/498 (20060101)

U.S. Cl.:

CPC H01L24/40 (20130101); H01L24/32 (20130101); H01L24/37 (20130101); H01L24/73 (20130101); H01L25/072 (20130101); H01L25/074 (20130101); H01L23/49833 (20130101); H01L23/49844 (20130101); H01L23/49861 (20130101); H01L24/48 (20130101); H01L2224/32225 (20130101); H01L2224/37124 (20130101); H01L2224/37147 (20130101); H01L2224/4001 (20130101); H01L2224/40091 (20130101); H01L2224/40095 (20130101); H01L2224/40137 (20130101); H01L2224/40145 (20130101); H01L2224/40225 (20130101); H01L2224/40245 (20130101); H01L2224/48225 (20130101); H01L2224/73213 (20130101); H01L2224/73215 (20130101); H01L2224/73221 (20130101); H01L2224/73263 (20130101); H01L2224/73265 (20130101); H01L2924/15724 (20130101); H01L2924/15747 (20130101); H01L2924/15787 (20130101); H01L2924/16196 (20130101); H01L2924/16251 (20130101); H01L2924/1632 (20130101); H01L2924/165 (20130101); H01L2924/182 (20130101); H01L2924/3512 (20130101); H01L2924/37001 (20130101)

Field of Classification Search

CPC: H01L (24/40)

USPC: 257/786

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Primary Examiner: Arora; Ajay

Attorney, Agent or Firm: Park, Kim & Suh, LLC

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims the benefit of Korean Patent Application No. 10-2021-0076510, filed on Jun. 14, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a clip structure for a semiconductor package and a semiconductor package including the same, wherein the clip structure, which is used instead of a high-priced metal spacer having a vertical structure formed of a material similar with a Coefficient of Thermal Expansion (CTE) of a semiconductor device so as to have price competitiveness, is bent having elasticity, and has a non-vertical structure, may disperse or absorb push-stress generated while molding so as to protect a semiconductor device and to effectively prevent a crack of an adhesive on a bonding area interposed between the semiconductor device and the clip structure.

Description of the Related Art

(3) In general, a semiconductor package includes a semiconductor chip installed on a lower substrate or an upper substrate, a conductor which is a metal post functioning as a spacer bonded on the semiconductor chip, a lead frame formed of Cu for applying an external electrical signal, and a package housing molded with a sealing member. Here, the semiconductor chip is bonded on a pad of the lead frame, and a plating layer formed of Ag is formed on a lead of the lead frame, wherein the plating layer is electrically connected to a pad of the semiconductor chip using a bonding wire which is a signal line.

(4) For example, as illustrated in FIG. 1A, in a general semiconductor package, a semiconductor chip **14** is bonded on a lower metal insulating substrate **11A** using a first bonding unit **12** interposed therebetween, a hexahedral or cylindrical conductor **17** having a vertical structure which is a metal spacer is bonded on the semiconductor chip **14** using a second bonding unit **16** interposed therebetween and is bonded on an upper metal insulating substrate **11B** using a third bonding unit **13** interposed therebetween, and a metal bridge **18** having a vertical structure is formed for electrical connection between the lower metal insulating substrate **11A** and the metal insulating substrate **11B**.

(5) The semiconductor chip is bonded to the substrate and the conductor by respectively using a solder interposed therebetween, however, a crack is generated in the first bonding unit **12** or the second bonding unit **16**, as illustrated in FIG. 1B, due to a difference of a Coefficient of Thermal Expansion (CTE) between the substrates **11A** and **11B**, the conductor **17**, and the first and second bonding units **12** and **16** and thereby, a reliability issue may occur.

(6) The main cause of the crack generated due to a CTE difference is direct impact on the semiconductor chip occurring after the metal spacer bonded on the semiconductor chip is directly and vertically bonded to the upper metal insulating substrate and thereby, a mold pressurizes the upper metal insulating substrate and the metal spacer during molding for forming a package housing. Accordingly, a yield of a product is lowered.

(7) In order to minimize a CTE difference with the semiconductor chip, the metal spacer or instead of a metal post, a material similar with a CTE of the semiconductor chip may be used, however, are considerably expensive compared with an existing metal spacer or metal post. Therefore, price competitiveness of a product is lowered.

SUMMARY OF THE INVENTION

(8) The present invention provides a clip structure for a semiconductor package and a

semiconductor package including the same, wherein the clip structure, which is used instead of a high-priced metal spacer having a vertical structure formed of a material similar with a Coefficient of Thermal Expansion (CTE) of a semiconductor device so as to have price competitiveness, is bent having elasticity, and has a non-vertical structure, may disperse or absorb push-stress generated while molding so as to protect a semiconductor device and to effectively prevent a crack of an adhesive on a bonding area interposed between the semiconductor device and the clip structure.

(9) According to an aspect of the present invention, there is provided a clip structure for a semiconductor package including: a first bonding unit bonded to a terminal part of an upper surface or a lower surface of a semiconductor device by using a conductive adhesive interposed therebetween; a main connecting unit which is extended and bent from the first bonding unit; a second bonding unit having an upper surface higher than the upper surface of the first bonding unit; an elastic unit elastically connected between the main connecting unit and one end of the second bonding unit; and a supporting unit bent and extended from the other end of the second bonding unit toward the main connecting unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit.

(10) The end of the supporting unit and the main connecting unit may be structurally contacted with each other.

(11) The end of the supporting unit and the main connecting unit may be spaced apart from each other by a distance of $0.1\ \mu\text{m}$ through $2\ \text{mm}$.

(12) The clip structure may be formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al and have a thickness of $0.1\ \text{mm}$ through $2\ \text{mm}$.

(13) The elastic unit may have a circular form.

(14) According to another aspect of the present invention, there is provided a semiconductor package including: at least one first substrate; at least one second substrate spaced apart from the first substrate; at least one semiconductor device installed on the first substrate or the second substrate; a clip structure including: a first bonding unit bonded to a terminal part of an upper surface or a lower surface of the semiconductor device by using a conductive adhesive interposed therebetween, a main connecting unit which is extended and bent from the first bonding unit, a second bonding unit having an upper surface higher than the upper surface of the first bonding unit, an elastic unit elastically connected between the main connecting unit and one end of the second bonding unit, and a supporting unit bent and extended from the other end of the second bonding unit toward the main connecting unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit; a package housing covering the semiconductor device; and at least one terminal exposed to the outside of the package housing for electrically connecting to the semiconductor device, wherein the second bonding unit of the clip structure is bonded to the second substrate, the first substrate, or the first and second substrates by using a conductive adhesive or a non-conductive adhesive.

(15) According to another aspect of the present invention, there is provided a semiconductor package including: at least one first substrate including at least one first semiconductor device installed thereon; at least one second substrate which is spaced apart from the first substrate, the at least one second substrate including at least one second semiconductor device installed thereon; a clip structure including: a first bonding unit bonded to a terminal part of an upper surface or a lower surface of the first semiconductor device by using a conductive adhesive interposed therebetween, a main connecting unit which is extended and bent from the first bonding unit, a second bonding unit having an upper surface higher than the upper surface of the first bonding unit, an elastic unit elastically connected between the main connecting unit and one end of the second bonding unit, and a supporting unit bent and extended from the other end of the second bonding unit toward the main connecting unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit; a package housing

covering the first semiconductor device and the second semiconductor device; and at least one terminal exposed to the outside of the package housing for electrically connecting to the first semiconductor device or the second semiconductor device, wherein the second bonding unit of the clip structure is bonded to a terminal part of the upper surface or the lower surface of the second semiconductor device by using a conductive adhesive.

(16) According to another aspect of the present invention, there is provided a semiconductor package including: at least one substrate; at least one semiconductor device installed on the substrate; a clip structure including: a first bonding unit bonded to a terminal part of an upper surface or a lower surface of the semiconductor device by using a conductive adhesive interposed therebetween, a main connecting unit which is extended and bent from the first bonding unit, a second bonding unit having an upper surface higher than the upper surface of the first bonding unit, an elastic unit elastically connected between the main connecting unit and one end of the second bonding unit, and a supporting unit bent and extended from the other end of the second bonding unit toward the main connecting unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit; a package housing covering the semiconductor device; and at least one terminal exposed to the outside of the package housing for electrically connecting to the semiconductor device, wherein at least a part of the upper surface of the second bonding unit in the clip structure is exposed to the surface of the package housing.

(17) The end of the supporting unit and the main connecting unit in the clip structure may be structurally contacted with each other.

(18) The end of the supporting unit and the main connecting unit in the clip structure may be spaced apart from each other by a distance of $0.1\text{ }\mu\text{m}$ through 2 mm .

(19) The clip structure may be formed of a single material comprising Cu or Al or a composite material containing 50% or more of Cu or Al and have a thickness of 0.1 mm through 2 mm .

(20) The elastic unit may have a circular form.

(21) The thickness of the clip structure may be smaller than the thickness of the first substrate or the second substrate.

(22) The substrate, the first substrate, or the second substrate may include at least one insulating layer.

(23) The substrate, the first substrate, or the second substrate may be formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al.

(24) At least a part of the second bonding unit exposed from the surface of the package housing may be bonded to a heat sink through a heat transmitting unit.

(25) The heat transmitting unit may have heat transfer rate of 0.1 w/mk through 15 w/m .

(26) The heat transmitting unit may be a conductor.

(27) A part of the substrate, the first substrate, or the second substrate may be exposed from the surface of the package housing.

(28) The semiconductor package described above may be used in driving an inverter, a converter, or an On Board Charger (OBC).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other features and advantages of the present invention will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings in which:

(2) FIG. 1 is a general semiconductor package;

(3) FIG. 2 is a perspective view of a clip structure for a semiconductor package according to a first

embodiment of the present invention;

(4) FIG. 3 is a cross-sectional view of the clip structure for a semiconductor package of FIG. 2;

(5) FIGS. 4 and 5 are cross-sectional views of a semiconductor package according to a second embodiment of the present invention;

(6) FIG. 6 is a cross-sectional view of a semiconductor package according to a third embodiment of the present invention;

(7) FIG. 7 is a cross-sectional view of a semiconductor package according to a fourth embodiment of the present invention;

(8) FIG. 8 illustrates a heat radiation structure of the semiconductor package according to the fourth embodiment of FIG. 7;

(9) FIG. 9 is a perspective view of the semiconductor package according to the second embodiment of FIG. 4;

(10) FIG. 10 is a side view and an exploded view of the semiconductor package of FIG. 9;

(11) FIG. 11 is an exploded perspective view of the semiconductor package of FIG. 9;

(12) FIG. 12 illustrates a design of a product where the semiconductor package of FIG. 9 is applied; and

(13) FIG. 13 is an exploded perspective view of the product of FIG. 12.

DETAILED DESCRIPTION OF THE INVENTION

(14) Hereinafter, embodiments of the present invention will be described in more detail with reference to accompanying drawings.

(15) A clip structure for a semiconductor package **110** according to a first embodiment of the present invention includes a first bonding unit **111** bonded to a terminal part of an upper surface or a lower surface of a semiconductor device **120** by using a conductive adhesive **110a** interposed therebetween, a main connecting unit **112** which is extended and bent from the first bonding unit **111**, a second bonding unit **113** having an upper surface higher than the upper surface of the first bonding unit **111**, an elastic unit **114** elastically connected between the main connecting unit **112** and one end of the second bonding unit **113**, and a supporting unit **115** bent and extended from the other end of the second bonding unit **113** toward the main connecting unit **112**, wherein the supporting unit **115** is formed to incline at an angle of 1° through 179° from an extended surface of the main connecting unit **112** and has an elastic structure so that push-stress applying to the semiconductor device while molding may be dispersed.

(16) Hereinafter, the clip structure for a semiconductor package **110** mentioned above will be described in more detail with reference to FIGS. 2 and 3.

(17) The clip structure **110** according to the present embodiment is bent in a specific form and thus, is elastic so that the clip structure **110** is bonded and electrically connected to the semiconductor device **120** and functions as a spacer which keeps a vertical bonding distance between a first substrate **130** and a second substrate **140**. Here, when the first substrate **130**, the second substrate **140**, and the semiconductor device **120** are molded, push-stress applied toward a vertical direction by a mold (not illustrated) may be dispersed to transmit the minimum impact on the semiconductor device **120**. Accordingly, the durability of the semiconductor device **120** may be improved.

(18) That is, the clip structure **110** may be classified by its structure and function and include the first bonding unit **111**, the main connecting unit **112**, the second bonding unit **113**, the elastic unit **114**, and the supporting unit **115**.

(19) More specifically, the clip structure **110** may include the planar type first bonding unit **111**, the main connecting unit **112**, the second bonding unit **113**, the elastic unit **114**, and the supporting unit **115**, wherein the first bonding unit **111** is bonded to the electrical terminal part of the upper surface (when installed on the first substrate **130**) or the lower surface (when installed on the second substrate **140**) of the semiconductor device **120** installed on the first substrate **130** or the second substrate **140** by using the conductive adhesive **110a** interposed therebetween, the main connecting unit **112** is extended and bent from the first bonding unit **111** to absorb the push-stress, the second

bonding unit **113** has the upper surface higher than the upper surface of the first bonding unit **111** to bond to the second substrate **140** or to be exposed to the surface of the package housing **150**, the elastic unit **114** is elastically connected between the main connecting unit **112** and one end of the second bonding unit **113** to absorb the push-stress, and the supporting unit **115** is bent downwards and extended from the other end of the second bonding unit **113** toward the main connecting unit **112**.

(20) As described above, the push-stress applied while molding for forming the package housing **150** is absorbed and dispersed left and right or upward and downward by using the elastic unit **114** elastically connected to the bent main connecting unit **112**. Accordingly, the semiconductor device **120** may be blocked from being directly impacted and thus, may have improved durability.

(21) In particular, the elastic unit **114** may have a circular form between the main connecting unit **112** and the second bonding unit **113** and thereby, the push-stress may be effectively dispersed in a vertical direction.

(22) Here, referring to FIG. 3A, the supporting unit **115** is formed to incline at an angle α of 1° through 179° from the extended surface of the main connecting unit **112**, and the push-stress applied in a vertical direction may be absorbed. Accordingly, the push-stress may not be directly transmitted to the semiconductor device **120**.

(23) Selectively, the end of the supporting unit **115** and the main connecting unit **112** may be structurally contacted with each other or may be spaced apart from each other by a distance d of $0.1\ \mu\text{m}$ through $2\ \text{mm}$, as illustrated in FIGS. 3A and 3B, according to a single sided substrate, a two-sided substrate, a material of the clip structure **110**, the semiconductor device, or a material of molding.

(24) In addition, the clip structure **110** may be generally formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al and may have a thickness of $0.1\ \text{mm}$ through $2\ \text{mm}$. Accordingly, the clip structure **110** may have excellent ductility and malleability and thus, may effectively absorb the push-stress.

(25) Hereinafter, semiconductor packages including the clip structure **110** described above will be described in more detail according to a substrate structure in the second through fourth embodiments.

(26) The semiconductor package according to the second embodiment of the present invention includes the first substrate **130**, the second substrate **140**, and the semiconductor device **120** contacting the first bonding unit **111**. The semiconductor package according to the third embodiment of the present invention includes the first substrate **130**, the second substrate **140**, and the semiconductor device **120** respectively contacting the first bonding unit **111** and the second bonding unit **113**. The semiconductor package according to the fourth embodiment of the present invention includes the single substrate **130**, the second bonding unit **113** exposed to the surface of the package housing **150**, and the semiconductor device **120** contacting the first bonding unit **111** or the main connecting unit **112**.

(27) Hereinafter, the semiconductor package according to the second embodiment of the present invention will be described in more detail below with reference to FIGS. 4 and 5.

(28) Firstly, at least one semiconductor device **120** is installed on at least one first substrate **130**, wherein the first substrate **130** may be a lower substrate having a specific metal pattern allowing electrical connection.

(29) Next, at least one second substrate **140** is spaced apart from the first substrate **130** and may be an upper substrate having a specific metal pattern allowing electrical connection. As illustrated in FIG. 4, the semiconductor device **120** may not be installed on the second substrate **140** and as illustrated in FIG. 5, the semiconductor device **120** may be installed on the second substrate **140**.

(30) Here, the first substrate **130** or the second substrate **140** may include at least one insulating layer including Al.sub.2O.sub.3 (ceramic) or AlN.

(31) Also, the first substrate **130** or the second substrate **140** may be formed of a metal material or

an insulating material.

(32) In addition, the first substrate **130** or the second substrate **140** may have a stack structure each including at least one metal layer **131** and **141**, at least one insulating layer **132** and **142** including Al.sub.2O.sub.3 (ceramic) or AlN, and at least one metal layer **133** and **143**, wherein each metal layer may have a thickness of 15 μ m through 2 mm and may be a single metal layer, an alloy metal layer, or a plated metal layer.

(33) The at least one semiconductor device **120** is installed on the first substrate **130** or the second substrate **140**, and one end thereof is bonded to the first substrate **130** (refer to FIG. 4) or the first and second substrates **130** and **140** (refer to FIG. 5) by using a conductive adhesive **121** interposed therebetween.

(34) The semiconductor device **120** is a power semiconductor chip such as an Insulated Gate Bipolar Transistor (IGBT), a diode, a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET), or a Junction Field Effect Transistor (JFET) and may be used in driving of an inverter, a converter, or an On Board Charger (OBC) which converts or controls power.

(35) Next, the clip structure for a semiconductor package **110** is mentioned above, and as illustrated in FIGS. 2 through 5, the clip structure **110** includes the first bonding unit **111** bonded to the terminal part of the upper surface or the lower surface of the semiconductor device **120** by using the conductive adhesive **110a** interposed therebetween, the main connecting unit **112** which is extended and bent from the first bonding unit **111**, the second bonding unit **113** having the upper surface higher than the upper surface of the first bonding unit **111**, the elastic unit **114** elastically connected between the main connecting unit **112** and one end of the second bonding unit **113**, and the supporting unit **115** bent and extended from the other end of the second bonding unit **113** toward the main connecting unit **112**, wherein the supporting unit **115** is formed to incline at an angle of 1° through 179° from the extended surface of the main connecting unit **112**. Accordingly, the push-stress may be dispersed or absorbed and thereby, the impact which may be directly transmitted to the semiconductor device **120** may be relieved.

(36) Here, the second bonding unit **113** of the clip structure **110** may be bonded to the second substrate **140**, the first substrate **130**, or the first and second substrates **130** and **140** by using a conductive adhesive **110b** or a non-conductive adhesive **110b** interposed therebetween.

(37) As illustrated in FIG. 4, the single clip structure **110** may be formed between the first substrate **130** and the second substrate **140** and as illustrated in FIG. 5, at least a pair of the single clip structure **110** may be formed and spaced apart from each other between the first substrate **130** and the second substrate **140** in a vertically reversed structure.

(38) Also, the main connecting unit **112** may be bonded to the first substrate **130** or the second substrate **140** by using a separate adhesive **110c**.

(39) A conductive connecting member (not illustrated) such as a bonding wire or a clip electrically connects the semiconductor device **120** to a terminal **160** extended to the outside of the package housing **150** and thus, an electrical signal may be applied (refer to FIGS. 7 and 8).

(40) The package housing **150** is formed of EMC, PBT, or PPS, insulates, and covers to protect an internal circuit, for example, the clip structure **110** and the semiconductor device **120**, and a part of the terminal **160** is exposed to the outside of the package housing **150**.

(41) At least one terminal **160** extends to be exposed to the outside of the package housing **150** for electrically connecting to the semiconductor device **120**.

(42) The clip structure **110** having an elastic structure may be bonded between the first substrate **130** and the second substrate **140** to function as a spacer and may be bent in a specific form to disperse or absorb the push-stress. Accordingly, the semiconductor device **120** may be protected from the outside impact.

(43) Hereinafter, the semiconductor package according to the third embodiment of the present invention will be described in more detail with reference to FIG. 6.

(44) Firstly, at least one first semiconductor device **120A** is installed on the at least one first

substrate **130**.

(45) Next, the at least one second substrate **140** is spaced apart from the first substrate **130** and includes at least one second semiconductor device **120B** installed thereon.

(46) The clip structure for a semiconductor package **110** is mentioned above, and as illustrated in FIGS. 2, 3, and 6, the clip structure **110** includes the first bonding unit **111** bonded to the terminal part of the upper surface or the lower surface of the first semiconductor device **120A** by using the conductive adhesive **110a** interposed therebetween, the main connecting unit **112** which is extended and bent from the first bonding unit **111**, the second bonding unit **113** having the upper surface higher than the upper surface of the first bonding unit **111**, the elastic unit **114** elastically connected between the main connecting unit **112** and one end of the second bonding unit **113**, and the supporting unit **115** bent and extended from the other end of the second bonding unit **113** toward the main connecting unit **112**, wherein the supporting unit **115** is formed to incline at an angle of 1° through 179° from the extended surface of the main connecting unit **112**. Accordingly, the push-stress may be dispersed or absorbed and thereby, the impact which may be directly transmitted to the semiconductor device **120** may be relieved.

(47) Here, the second bonding unit **113** of the clip structure **110** may be bonded to the terminal part of the upper surface or the lower surface of the second semiconductor device **120B** by using the conductive adhesive **110a** interposed therebetween.

(48) A conductive connecting member (not illustrated) such as a bonding wire or a clip electrically connects the first semiconductor device **120A** or the second semiconductor device **120B** to the terminal **160** extended to the outside of the package housing **150** and thus, an electrical signal may be applied (refer to FIGS. 7 and 8).

(49) The package housing **150** is formed of EMC, PBT, or PPS, insulates, and covers to protect an internal circuit, for example, the clip structure **110**, the first semiconductor device **120A**, and the second semiconductor device **120B**, and a part of the terminal **160** is exposed to the outside of the package housing **150**.

(50) At least one terminal **160** extends to be exposed to the outside of the package housing **150** for electrically connecting to the first semiconductor device **120A** or the second semiconductor device **120B**.

(51) The clip structure **110** having an elastic structure may be bonded between the first substrate **130** and the second substrate **140** to function as a spacer and may be bent in a specific form to disperse or absorb the push-stress. Accordingly, the first semiconductor device **120A** and the second semiconductor device **120B** may be protected from the outside impact.

(52) Hereinafter, the semiconductor package according to the fourth embodiment of the present invention will be described in more detail with reference to FIGS. 7 and 8.

(53) Firstly, at least one semiconductor device **120** is installed on the at least one substrate **130**, wherein the substrate **130** may be a lower substrate having a specific metal pattern allowing electrical connection.

(54) Next, the clip structure for a semiconductor package **110** is mentioned above, and as illustrated in FIGS. 2, 3, 7, and 8, the clip structure **110** includes the first bonding unit **111** bonded to the terminal part of the upper surface or the lower surface of the semiconductor device **120** by using the conductive adhesive **110a** interposed therebetween, the main connecting unit **112** which is extended and bent from the first bonding unit **111**, the second bonding unit **113** having the upper surface higher than the upper surface of the first bonding unit **111**, the elastic unit **114** elastically connected between the main connecting unit **112** and one end of the second bonding unit **113**, and the supporting unit **115** bent and extended from the other end of the second bonding unit **113** toward the main connecting unit **112**, wherein the supporting unit **115** is formed to incline at an angle of 1° through 179° from the extended surface of the main connecting unit **112**. Accordingly, the push-stress may be dispersed or absorbed and thereby, the impact which may be directly transmitted to the semiconductor device **120** may be relieved.

(55) Here, at least a part of the upper surface of the second bonding unit **113** in the clip structure **110** is exposed to the surface of the package housing **150** so that heat generated while driving the semiconductor device **120** may be transmitted to the outside of the package housing **150** through the clip structure **110** and radiated.

(56) Also, as illustrated in FIG. 7A, the main connecting unit **112** may be bonded to the substrate **130** by using the separate adhesive **110c** and as illustrated in FIG. 7B, the main connecting unit **112** may be bonded to the substrate **130** by using the adhesive **110a** formed on the upper surface of the other semiconductor device **120** on the substrate **130**.

(57) A conductive connecting member such as a bonding wire or a clip electrically connects the semiconductor device **120** to the terminal **160** extended to the outside of the package housing **150** and thus, an electrical signal may be applied.

(58) The package housing **150** is formed of EMC, PBT, or PPS, insulates, and covers to protect an internal circuit, for example, the clip structure **110** and the semiconductor device **120**, and a part of the terminal **160** is exposed to the outside of the package housing **150**.

(59) At least one terminal **160** extends to be exposed to the outside of the package housing **150** for electrically connecting to the semiconductor device **120**.

(60) The clip structure **110** having an elastic structure may be bent in a specific form to disperse or absorb the push-stress. Accordingly, the semiconductor device **120** may be protected from the outside impact and a part of the clip structure **110** is exposed to the surface of the package housing **150** so that heat may be radiated.

(61) Commonly in the semiconductor packages of the second through fourth embodiments, the end of the supporting unit **115** and the main connecting unit **112** may be structurally contacted with each other as illustrated in FIG. 3A or may be spaced apart from each other by a distance d of 0.1 μm through 2 mm as illustrated in FIG. 3B.

(62) Also, the clip structure **110** may be generally formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al and may have a thickness of 0.1 mm through 2 mm. Accordingly, the clip structure **110** may have excellent ductility and malleability and thus, may effectively absorb the push-stress.

(63) In addition, the elastic unit **114** may have a circular form between the main connecting unit **112** and the second bonding unit **113** and thereby, the push-stress may be effectively dispersed in a vertical direction.

(64) The thickness of the clip structure **110** may be smaller than the thicknesses of the substrate **130** (fourth embodiment), the first substrate **130** (second embodiment and third embodiment), or the second substrate **140** (second embodiment and third embodiment).

(65) The substrate **130**, the first substrate **130**, or the second substrate **140** may include at least one insulating layer including Al.sub.2O.sub.3 (ceramic) or AlN.

(66) Also, the substrate **130**, the first substrate **130**, or the second substrate **140** may be formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al. For example, as illustrated in FIG. 7B, the substrate **130** may be a metal substrate formed of a single material including Cu or Al or a composite material containing 50% or more of Cu or Al.

(67) The terminal **160** may be bonded to the substrate **130**, the first substrate **130** or the second substrate **140** by using a conductive adhesive or may be electrically connected to the substrate **130**, the first substrate **130** or the second substrate **140** by ultrasonic welding.

(68) Referring to FIG. 8, at least a part of the second bonding unit **113** exposed from the surface of the package housing **150** is bonded to a heat sink **172** through a heat transmitting unit **171** so that heat generated from the clip structure **110** to the heat sink **172** through the heat transmitting unit **171** may be effectively radiated to the outside of the package housing **150**.

(69) Here, the heat sink **172** may include a plurality of cooling fins and may cool the heat by air-cooling or water-cooling.

(70) Also, the heat transmitting unit **171** may be a conductor having heat transfer rate of 0.1 w/mk

through 15 w/m.

(71) The substrate **130** (refer to FIGS. 7 and 8), a part of the first substrate **130** (refer to FIGS. 4 through 6), or the second substrate **140** (refer to FIGS. 4 through 6) are exposed from the surface of the package housing **150** and thus, heat may be radiated through the substrates exposed from the package housing **150**.

(72) The semiconductor packages described above may be used in driving an inverter, a converter, or an OBC which converts or controls power.

(73) FIG. 9 is a perspective view of the semiconductor package according to the second embodiment of FIG. 4, FIG. 10 is a side view and an exploded view of the semiconductor package of FIG. 9, FIG. 11 is an exploded perspective view of the semiconductor package of FIG. 9, FIG. 12 illustrates a design of a product where the semiconductor package of FIG. 9 is applied, and FIG. 13 is an exploded perspective view of the product of FIG. 12.

(74) Referring to the drawings above, the clip structure **110** having an elastic structure and a thickness of 0.2 mm is formed between the first and second substrates **130** and **140** each of which is formed of Zirconia Toughened Alumina (ZTA) having a thickness of 0.3 mm, instead of a general metal spacer having a vertical structure. The first bonding unit **111** is bonded to the semiconductor device **120** by using the conductive adhesive **110a** formed of a Sn solder material, the main connecting unit **112** is bonded to the first substrate **130** by using the adhesive **110c** formed of a Sn solder material, and the second bonding unit **113** is bonded to the second substrate **140** by using the conductive adhesive **110b** formed of a SAC305 (Sn 96.5%, Ag 3.0%, Cu 0.5%) solder material. Accordingly, while molding, the push-stress applied between the first substrate **130** and second substrate **140** may not be directly transmitted to the semiconductor device **120**.

(75) FIGS. 12 and 13 illustrate a lead frame **180** where the terminal **160** electrically connected to the first substrate **130** or the second substrate **140** is patterned. In the semiconductor package described above, a part of the terminal **160** is extended and exposed. Then, molding is performed to form the package housing **150**. Here, the lead frame **180** disposed outside of the package housing **150** is cut for commercialization.

(76) According to the clip structure for the semiconductor package and the semiconductor package including the same described above, the clip structure, which is used instead of a high-priced metal spacer having a vertical structure formed of a material similar with a CTE of the semiconductor device so as to have price competitiveness, is bent having elasticity, and has a non-vertical structure, may disperse or absorb push-stress generated while molding so as to protect the semiconductor device and to effectively prevent a crack of an adhesive on a bonding area interposed between the semiconductor device and the clip structure, and may expose a part of the substrate or the clip structure to the outside of the package housing so as to improve heat radiation efficiency.

(77) According to the present invention, the clip structure, which is used instead of a high-priced metal spacer having a vertical structure formed of a material similar with a CTE of the semiconductor device so as to have price competitiveness, is bent having elasticity, and has a non-vertical structure, may disperse or absorb push-stress generated while molding so as to protect the semiconductor device and to effectively prevent a crack of an adhesive on a bonding area interposed between the semiconductor device and the clip structure, and may expose a part of the substrate or the clip structure to the outside of the package housing so as to improve heat radiation efficiency. While the present invention has been particularly shown and described with reference to exemplary embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present invention as defined by the following claims.

(78) TABLE-US-00001 [Reference Signs and Numerals] 110: clip structure 111: a first bonding unit 112: a main connecting unit 113: a second bonding unit 114: an elastic unit 115: a supporting unit 120: a semiconductor device 120A: a first semiconductor device 120B: a second

semiconductor device 121: a conductive adhesive 130: a substrate, a first substrate 131: a metal layer 132: a insulating layer 133: a metal layer 140: a second substrate 141: a metal layer 142: a insulating layer 143: a metal layer 150: a package housing 160: a terminal 171: a heat transmitting unit 172: a heat sink 180: a lead frame

Claims

1. A clip structure for a semiconductor package comprising: a first lower bonding unit configured to be bonded to a terminal part of an upper surface of a semiconductor device formed on a lower substrate; a first connecting unit folded upward extending from an end of the first lower bonding unit; a second connecting unit extending from an end of the first connecting unit in a horizontal direction; a third connecting unit folded downward extending from an end of the second connecting unit; a second lower bonding unit extending from an end of the third connecting unit in the horizontal direction and configured to be bonded to the lower substrate; an upper bonding unit provided above the second lower bonding unit and configured to be bonded to an upper substrate; and an elastic unit elastically connected between a first end of the upper bonding unit and an end of the second lower bonding unit, the elastic unit having an arch shape.
2. The clip structure for a semiconductor package of claim 1, further comprising a supporting unit folded downward extending from a second end of the upper bonding unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from the second lower bonding unit.
3. The clip structure for a semiconductor package of claim 2, wherein an end of the supporting unit is in contact with the second lower bonding unit.
4. The clip structure for a semiconductor package of claim 2, wherein an end of the supporting unit and the second lower bonding unit are spaced apart from each other by a vertical distance of 0.1 μm through 2 mm.
5. The clip structure for a semiconductor package of claim 1, wherein the clip structure is formed of a single material comprising Cu or Al or a composite material containing 50% or more of Cu or Al and has a thickness of 0.1 mm through 2 mm.
6. The clip structure for a semiconductor package of claim 1, wherein the elastic unit has a semi-circular shape.
7. A semiconductor package comprising: a lower substrate; an upper substrate spaced apart from the lower substrate; a semiconductor device formed on the lower substrate; a clip structure comprising: a first lower bonding unit bonded to an upper surface of the semiconductor device; a first connecting unit folded upward extending from an end of the first lower bonding unit; a second connecting unit extending from an end of the first connecting unit in a horizontal direction; a third connecting unit folded downward extending from an end of the second connecting unit; a second lower bonding unit extending from an end of the third connecting unit in the horizontal direction and bonded to the lower substrate; an upper bonding unit provided above the second lower bonding unit and bonded to the upper substrate; and an elastic unit elastically connected between a first end of the upper bonding unit and an end of the second lower bonding unit, the elastic unit having an arch shape; a package housing covering the semiconductor device; and a terminal exposed to an outside of the package housing for electrically connecting to the semiconductor device.
8. A semiconductor package comprising: a lower substrate on which a first semiconductor device is formed; an upper substrate spaced apart from the lower substrate, wherein a second semiconductor device is formed on the upper substrate; a clip structure comprising: a first lower bonding unit bonded to an upper surface of the first semiconductor device; a first connecting unit folded upward extending from an end of the first lower bonding unit; a second connecting unit extending from an end of the first connecting unit in a horizontal direction; a third connecting unit folded downward extending from an end of the second connecting unit; a second lower bonding unit extending from an end of the third connecting unit in the horizontal direction and bonded to the lower substrate; an

upper bonding unit provided above the second lower bonding unit and bonded to a lower surface of the second semiconductor device; and an elastic unit elastically connected between a first end of the upper bonding unit and an end of the second lower bonding unit, the elastic unit having an arch shape; a package housing covering the first semiconductor device and the second semiconductor device; and a terminal exposed to an outside of the package housing for electrically connecting to the first semiconductor device or the second semiconductor device.

9. A semiconductor package comprising: a lower substrate; a semiconductor device formed on the lower substrate; a clip structure comprising: a first lower bonding unit bonded to an upper surface of the semiconductor device; a first connecting unit folded upward and extending from an end of the first lower bonding unit; a second connecting unit extending from an end of the first connecting unit in a horizontal direction; a third connecting unit folded downward extending from an end of the second connecting unit; a second lower bonding unit extending from an end of the third connecting unit in the horizontal direction and bonded to the lower substrate; an upper bonding unit provided above the second lower bonding unit; and an elastic unit elastically connected between a first end of the upper bonding unit and an end of the second lower bonding unit, the elastic unit having an arch shape; a package housing covering the semiconductor device; and a terminal exposed to an outside of the package housing for electrically connecting to the semiconductor device, wherein at least a part of an upper surface of the upper bonding unit in the clip structure is not covered with the package housing.

10. The semiconductor package of claim 7, wherein the clip structure further comprises a supporting unit folded downward extending from a second end of the upper bonding unit, wherein the supporting unit is formed to incline at an angle of 1° through 179° from the second lower bonding unit.

11. The semiconductor package of claim 10, wherein an end of the supporting unit is in contact with the second lower bonding unit.

12. The semiconductor package of claim 10, wherein an end of the supporting unit and the second lower bonding unit are spaced apart from each other by a vertical distance of $0.1\text{ }\mu\text{m}$ through 2 mm .

13. The semiconductor package of claim 7, wherein the clip structure is formed of a single material comprising Cu or Al or a composite material containing 50% or more of Cu or Al and has a thickness of 0.1 mm through 2 mm .

14. The semiconductor package of claim 7, wherein the elastic unit has a semi-circular shape.

15. The semiconductor package of claim 7, wherein a thickness of the clip structure is less than a thickness of the lower substrate or the upper substrate.

16. The semiconductor package of claim 7, wherein the lower substrate or the upper substrate comprises at least one insulating layer.

17. The semiconductor package of claim 7, wherein the lower substrate or the upper substrate is formed of a single material comprising Cu or Al or a composite material containing 50% or more of Cu or Al.

18. The semiconductor package of claim 7, wherein a part of the lower substrate or the upper substrate is not covered with the package housing.

19. The semiconductor package of claim 9, wherein the upper surface of the upper bonding unit is bonded to a heat sink through a heat transmitting unit.

20. A semiconductor package comprising the semiconductor package of claim 7 used in driving any one of an inverter, a converter, and an On Board Charger (OBC).
